



Hello from Docker Flask App!

f54aa13f80e6 ubuntu "bash" 13 days ago Up 10 hours

basit@SYEDBASIT26 MINGW64 ~/downloads/docker-labs/project-01-python-flask-docker (main)

```
$ docker exec -it flask-app bash
```

```
root@0987ae9661de:/app# ls
```

```
Dockerfile  README.md  app.py  requirements.txt
```

```
root@0987ae9661de:/app# |
```

```
basit@SYED8ASIT26 MINGW64 ~/downloads/docker-labs/project-01-python-flask-docker (main)
$ nano Dockerfile

basit@SYED8ASIT26 MINGW64 ~/downloads/docker-labs/project-01-python-flask-docker (main)
$ docker built -t flask-demo .
unknown shorthand flag: 't' in -t

Usage:  docker [OPTIONS] COMMAND [ARG...]

Run 'docker --help' for more information

basit@SYED8ASIT26 MINGW64 ~/downloads/docker-labs/project-01-python-flask-docker (main)
$ docker build -t flask-demo .
[+] Building 22.9s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 974B
=> [internal] load metadata for docker.io/library/python:3.12
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:3.12sha256:4037ca81380292f2eac4fc4438e55a29046b7dae5d8b53b6e1c41036d7cc98b6
=> => resolve docker.io/library/python:3.12sha256:4037ca81380292f2eac4fc4438e55a29046b7dae5d8b53b6e1c41036d7cc98b6
=> => sha256:d04536ac5ef055751b60fa87af5497578de31c633adeb26484adee16ae7ef4c5 247B / 247B
=> => sha256:d57bbc53353ecabb779b6c8f9b0af53272370a6ad158b763c050a8afab508f84 25.72MB / 25.72MB
=> => sha256:71eb69eeaf0140423be38ed241ad5ddc36be392df1205a10bc5c976ee3ef4e8b 6.09MB / 6.09MB
=> => sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 67.78MB / 67.78MB
=> => sha256:442ccal22a8b0987d6ea29e5fa7d0b8caf699cb127ec5fdbd8191faf86da521f 236.20MB / 236.20MB
=> => sha256:ef1b08ddd59d42b444e8f9c68a6cebbd70636aed0c242fdccb0bd61b61ba26b 49.32MB / 49.32MB
=> => sha256:8b365b6ff7b7e310e1b9a88f996e65b89969c7fa450492b68249d1d1c38e0676 25.64MB / 25.64MB
=> => extracting sha256:ef1b08ddd59d42b444e8f9c68a6cebbd70636aed0c242fdccb0bd61b61ba26b 2.2s
=> => extracting sha256:8b365b6ff7b7e310e1b9a88f996e65b89969c7fa450492b68249d1d1c38e0676 0.9s
=> => extracting sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 3.0s
=> => extracting sha256:442ccal22a8b0987d6ea29e5fa7d0b8caf699cb127ec5fdbd8191faf86da521f 4.9s
=> => extracting sha256:71eb69eeaf0140423be38ed241ad5ddc36be392df1205a10bc5c976ee3ef4e8b 0.2s
=> => extracting sha256:d57bbc53353ecabb779b6c8f9b0af53272370a6ad158b763c050a8afab508f84 0.6s
=> => extracting sha256:d04536ac5ef055751b60fa87af5497578de31c633adeb26484adee16ae7ef4c5 0.0s
=> [internal] load build context
=> => transferring context: 1.28kB
=> [2/5] WORKDIR /app
=> [3/5] COPY requirements.txt .
=> [4/5] RUN pip install -r requirements.txt
=> [5/5] COPY . .
=> => exporting to image
=> => exporting layers
=> => exporting manifest sha256:fbcb067c193b4b489e95dc80e379e3ec02c77d447c84c6d42b758584724bd0f5
=> => exporting config sha256:18f967afb5f09fc7716652d9d781b6a7e26f18d1ba90d13b0052565f943e99
=> => exporting attestation manifest sha256:6e8668439b4b0bc5ddaa572cd0a62efa089d329e56632263f116f568990e521a
=> => exporting manifest list sha256:a9aff78f5aa2eb74b1e5fbf06fb9cf81285fed39a1ee8f4cd53bb9483a11fe9
=> => naming to docker.io/library/flask-demo:latest
=> => unpacking to docker.io/library/flask-demo:latest
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/9hoibuu84nhga6x5wka9v161e

basit@SYED8ASIT26 MINGW64 ~/downloads/docker-labs/project-01-python-flask-docker (main)
$ docker run -d --name flask-app -p 5000:5000 flask-demo
0987ae9661de1d13b64e9e9e30dfca5fc47be5d203900b9b294db334bf4b018

basit@SYED8ASIT26 MINGW64 ~/downloads/docker-labs/project-01-python-flask-docker (main)
$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
0987ae9661de   flask-demo    "python app.py"         9 seconds ago Up 9 seconds  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   flask-app
11510213840e   devboard-frontend "docker-entrypoint.s..." 54 minutes ago Up 54 minutes  0.0.0.0:5173->5173/tcp, [::]:5173->5173/tcp   serene_goodall
a1b87f368b54   nginx         "/docker-entrypoint..." 31 hours ago  Up 6 hours    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp       web-app
f54aa13f80e6   ubuntu       "bash"                  13 days ago   Up 10 hours                               myubuntu

basit@SYED8ASIT26 MINGW64 ~/downloads/docker-labs/project-01-python-flask-docker (main)
$ |
```

```
$ nano Dockerfile
```

```
basit@SYEDBASIT26 MINGW64 ~/downloads/docker-labs/project-01-python-flask-docker (main)
```

```
$ docker built -t flask-demo .
```

```
unknown shorthand flag: 't' in -t
```

```
Usage: docker [OPTIONS] COMMAND [ARG...]
```

```
Run 'docker --help' for more information
```

```
basit@SYEDBASIT26 MINGW64 ~/downloads/docker-labs/project-01-python-flask-docker (main)
```

```
$ docker build -t flask-demo .
```

```
[+] Building 7.9s (5/10)
```

```
=> [internal] load build definition from Dockerfile
```

```
=> => transferring dockerfile: 974B
```

```
=> [internal] load metadata for docker.io/library/python:3.12
```

```
=> [auth] library/python:pull token for registry-1.docker.io
```

```
=> [internal] load .dockerignore
```

```
=> => transferring context: 2B
```

```
=> [1/5] FROM docker.io/library/python:3.12@sha256:4037ca81380292f2eac4fc4438e55a29046b7dae5d8b53b6e1c41036d7cc98b6
```

```
=> => resolve docker.io/library/python:3.12@sha256:4037ca81380292f2eac4fc4438e55a29046b7dae5d8b53b6e1c41036d7cc98b6
```

```
=> => sha256:d04536ac5ef055751b60fa87af5497578de31c633adeb26484adae16ae7ef4c5 247B / 247B
```

```
=> => sha256:d57bbc53353ecabb779b6c8f9b0af53272370a6ad158b763c050a8afab508f84 25.72MB / 25.72MB
```

```
=> => sha256:71eb69eeaf0140423be38ed241ad5ddc36be392df1205a10bc5c976ee3ef4e8b 6.09MB / 6.09MB
```

```
=> => sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 67.78MB / 67.78MB
```

```
=> => sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fdbd8191faf86da521f 121.63MB / 236.20MB
```

```
=> => sha256:ef1b08ddd59d42b444e8f9c68a6cebbbed70636aed0c242fdccb0bd61b61ba26b 49.32MB / 49.32MB
```

```
=> => sha256:8b365b6ff7b7e310e1b9a88f996e65b89969c7fa450492b68249d1d1c38e0676 25.64MB / 25.64MB
```

```
=> => extracting sha256:ef1b08ddd59d42b444e8f9c68a6cebbbed70636aed0c242fdccb0bd61b61ba26b
```

```
=> [internal] load build context
```

```
=> => transferring context: 1.28kB
```



```
#from python since this is a pythin app
FROM python:3.12
```

```
#create work directory where this code stays, every command after this runs inside app folder
WORKDIR /app
```

```
#copy the code bascially source to destination so copy req.txt to . mean current directory inside app folder
COPY requirements.txt .
```

```
#install required dependencies
RUN pip install -r requirements.txt
```

```
#copy the application files
COPY . .
```

```
#expose port, tell application to listen on port
EXPOSE 5000
```

```
# to run this image
CMD ["python", "app.py"]
```

```
#FROM python:3.12
```

```
#      |
#Create Linux + Python Image
```

```
#      |
#WORKDIR /app
```

```
#      |
#Move into /app
```

```
#      |
#COPY requirements.txt .
```

```
#      |
#Copy dependency list
```

```
#      |
#RUN pip install -r requirements.txt
```

```
#      |
#Install Flask
```

```
#      |
#COPY . .
```

```
#      |
#Copy application files
```

```
#      |
#EXPOSE 5000
```

```
#      |
#Application uses port 5000
```

```
#      |
#CMD ["python","app.py"]
```

```
#      |
#Start Flask app
```

```
#from python since this is a pythin app
FROM python:3.12

#create work directory where this code stays
WORKDIR /app

#copy the code
COPY requirements.txt .

#install required dependencies
RUN pip install -r requirements.txt

...
COPY . .

EXPOSE 5000

CMD ["python", "app.py"]
```



```
basit@SYEDBASIT26 MINGW64 ~/downloads/node/devboard (master)
$ cd ..

basit@SYEDBASIT26 MINGW64 ~/downloads/node
$ cd ..

basit@SYEDBASIT26 MINGW64 ~/downloads
$ git clone git@github.com:saftab4-arch/docker-labs.git
Cloning into 'docker-labs'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
Resolving deltas: 100% (1/1), done.

basit@SYEDBASIT26 MINGW64 ~/downloads
$ cd docker-labs

basit@SYEDBASIT26 MINGW64 ~/downloads/docker-labs (main)
$ mkdir project-01-python-flask-docker

cd project-01-python-flask-docker

basit@SYEDBASIT26 MINGW64 ~/downloads/docker-labs/project-01-python-flask-docker (main)
$ ls

basit@SYEDBASIT26 MINGW64 ~/downloads/docker-labs/project-01-python-flask-docker (main)
$ touch app.py

touch requirements.txt

touch Dockerfile

touch README.md

basit@SYEDBASIT26 MINGW64 ~/downloads/docker-labs/project-01-python-flask-docker (main)
$ ls
Dockerfile  README.md  app.py  requirements.txt

basit@SYEDBASIT26 MINGW64 ~/downloads/docker-labs/project-01-python-flask-docker (main)
$ |
```

